

Sustained virologic suppression of multidrug-resistant HIV in an individual treated with anti-CD4 domain 1 antibody and lenacapavir.

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The clinical management of people with multidrug-resistant (MDR) human immunodeficiency virus (HIV) remains challenging despite continued development of antiretroviral agents. A 58-year-old male individual with MDR HIV and Kaposi sarcoma (KS) was treated with a new antiretroviral regimen consisting of anti-CD4 domain 1 antibody UB-421 and capsid inhibitor lenacapavir. The individual experienced delayed but sustained suppression of plasma viremia and a substantial increase in the CD4+ T cell count. A longitudinal examination of plasma HIV and infectious isolates showed no evidence of viral evolution or the emergence of UB-421- or lenacapavir-resistant viruses. The individual received three cycles of liposomal doxorubicin and five doses of anti-programmed cell death protein 1 (PD-1) monoclonal antibody pembrolizumab that resulted in improvement in KS with flattening of lesions. Our data demonstrate that combination therapy with UB-421 could provide sustained virologic suppression in people harboring MDR HIV with limited therapeutic alternatives.

Lenacapavir: A Novel Long-Acting Capsid Inhibitor for HIV.

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To review the efficacy, safety, and role of lenacapavir (LEN) in the treatment of HIV-1 infection. A literature search was performed using PubMed and Google Scholar (through March 2023) with the search term LEN and GS-6207. Other resources included abstracts presented at recent conferences, the manufacturer's Web site, and prescribing information. All relevant articles, trial updates, and conference abstracts in the English language were included. Lenacapavir represents a new class of antiretrovirals (ARVs) with a novel mechanism of action as a capsid inhibitor and a unique twice-a-year subcutaneous administration schedule. Lenacapavir when combined with other ARVs has proven to benefit heavily treatment-experienced (HTE) patients with HIV-1 infection in achieving viral suppression and immune restoration. Lenacapavir is a new treatment option that patients who are HTE can consider adding as part of an ARV regimen. Lenacapavir is an effective and well-tolerated option for HTE patients which is a valuable addition to the arsenal of ARVs.

Lenacapavir: a first-in-class HIV-1 capsid inhibitor.

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This review summarizes available data for lenacapavir, an investigational first-in-class agent that disrupts functioning of HIV capsid protein across multiple steps in the viral life cycle. Lenacapavir demonstrated picomolar potency in vitro with no cross resistance to existing antiretroviral classes and potent antiviral activity in persons with HIV-1. In persons with HIV-1, there was no preexisting resistance to lenacapavir regardless of treatment history. Lenacapavir can be administered orally either daily or weekly and subcutaneously up to every 6 months. In heavily treatment-experienced persons with multidrug-resistant HIV-1 and in treatment-naïve persons with HIV-1, lenacapavir in combination with other antiretroviral agents led to high rates of virologic suppression and was well tolerated. Ongoing studies are evaluating long-acting dosing of lenacapavir for treating HIV-1 in combination with other antiretrovirals and

preventing HIV-1 as a single agent.

Lenacapavir: a twice-yearly treatment for adults with multidrug-resistant HIV infection and limited treatment options.

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Lenacapavir, the compound reviewed in this paper, is a novel, potent, and long-acting first-in-class HIV-1 capsid inhibitor that was granted a breakthrough therapy designation for HIV treatment by the U.S. Food and Drug Administration in May 2019. Lenacapavir was subsequently approved, in combination with other antiretroviral agents, for the treatment of multidrug-resistant HIV in people with HIV who have limited treatment options due to safety, intolerance, and resistance considerations, based on clinical trial data, which demonstrated its efficacy, tolerability, and safety for that population. This review draws upon published and unpublished data of lenacapavir (GS-6207) to discuss its chemistry, mechanism of action, pharmacokinetic, and pharmacodynamic properties, as well as efficacy and safety observed in clinical trials; and an expert opinion section discusses its clinical uses, advantages, limitations, and potential future applications. Lenacapavir should be used in combination with at least one or two additional fully active agents, where possible, and adherence support is critical for selected companion drugs (particularly if self-administered) to optimize efficacy and prevent treatment-emergent resistance to lenacapavir. Providers should pay attention to drug-drug interactions when initiating lenacapavir.

Lenacapavir: A first-in-class capsid inhibitor for the treatment of highly treatment-resistant HIV.

PMID: 37767713 | Indexed in PubMed

The purpose of this article is to review the pharmacology, efficacy, and safety of the capsid inhibitor lenacapavir for the treatment of multidrug-resistant human immunodeficiency virus type 1 (HIV-1) infection. A review of the literature was performed by searching PubMed/MEDLINE for all relevant articles published between February 2021 and March 2023 using the keywords "lenacapavir," "Sunlenca," "human immunodeficiency virus," and "treatment" together with "multidrug resistant human immunodeficiency virus." All English-language articles describing clinical trials assessing the efficacy and safety of lenacapavir when used in humans for the treatment of HIV infection were included. Review articles, conference abstracts, and article references were evaluated for relevant information, and data were also obtained from the manufacturer's website and the package insert. Lenacapavir has been approved by the Food and Drug Administration (FDA) for the treatment of HIV-1 infection in heavily treatment-experienced adults with multidrug resistance for whom the current antiretroviral regimen is failing due to resistance, intolerance, or safety considerations. It is the first in a new class of drugs called capsid inhibitors to receive FDA approval. Lenacapavir is a long-acting subcutaneous injectable to be administered once every 6 months. The phase 3 clinical trial evaluating lenacapavir has demonstrated its efficacy in viral load reduction from baseline compared to placebo in patients receiving optimized background therapy. The most common adverse events reported in the clinical trial were injection site reactions, occurring in 63% of participants. Lenacapavir is a novel capsid inhibitor indicated, in combination with other antiretroviral therapy, for treatment of multidrug-resistant HIV-1 infection.

Lenacapavir: First Approval.

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Lenacapavir (Sunlenca®) is a long-acting capsid inhibitor of human immunodeficiency virus type 1 (HIV-1) being developed by Gilead Sciences Inc. It is available as an oral tablet and injectable solution, with the latter being a slow-release formulation to allow bi-annual subcutaneous administration. In August 2022, lenacapavir received its first approval in the EU for use in combination with other antiretroviral(s) in adults with multi-drug resistant HIV infection, for whom it is otherwise not possible to construct a suppressive anti-viral regimen. This article summarizes the milestones in the development of lenacapavir leading to this first approval for the treatment of HIV-1 infection.

Lenacapavir: A novel injectable HIV-1 capsid inhibitor.

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Patients living with multidrug-resistant (MDR) HIV have limited antiretroviral regimen options that provide durable viral suppression. Lenacapavir is a novel first-in-class inhibitor of HIV-1 capsid function with efficacy at various stages of the viral life cycle, and it is indicated for the treatment of MDR HIV-1 infection in combination with optimized background antiretroviral therapy. The favourable pharmacokinetic profile supports an every sixth month dosing interval of subcutaneous lenacapavir after an initial oral loading dose, which may advocate for continued adherence to antiretroviral therapy (ART) through the reduction of daily pill burden. The role of lenacapavir in promoting virologic suppression has been studied in patients with MDR HIV-1 on failing ART at baseline. Lenacapavir was well tolerated in clinical trials with the most common adverse effects including mild to moderate injection site reactions, gastrointestinal symptoms, and headache. Substitutions on the capsid molecule may confer resistance to lenacapavir by changing the binding potential. Cross-resistance to other antiretrovirals has not been observed. The unique mechanism of action, pharmacokinetics, and safety and efficacy of lenacapavir support its use for the management of MDR HIV-1 infection. Current studies are ongoing to evaluate the potential use of subcutaneous lenacapavir for pre-exposure prophylaxis (PrEP). Future studies will confirm the long-term clinical safety, efficacy, and resistance data for lenacapavir.

Lenacapavir and the novel HIV-1 capsid inhibitors: an emerging therapy in the management of multidrug-resistant HIV-1 virus.

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The aim of this study was to summarize current evidence regarding lenacapavir, a first in class HIV-1 capsid inhibitor, and its role as an emergent therapy for the treatment of HIV-1 infection. HIV-1 capsid inhibitors (of which lenacapavir is the first in class) has been postulated to have activity against multidrug resistant HIV-1 viral isolates. Initial results from the phase 3 trial CAPELLA (combining oral and subcutaneous lenacapavir alongside failing drug therapies) suggest that there may be a role for these novel agents in a cohort of patients living with HIV-1 infection (PLWH) for whom multidrug resistance has previously been a barrier to effective therapy. Despite emergent lenacapavir resistance mutations detected in some study participants, virological suppression was still potentially attainable, offering some hope to PLWH with limited antiviral regimens available. Initial results from the CALIBRATE trial show

promise for the role of lenacapavir-containing regimens in a treatment-naïve cohort as well. Lenacapavir may prove to be an adjunctive agent in the management of PLWH with significant HIV-1 drug resistance.

Efficacy and Safety of Long-Acting Subcutaneous Lenacapavir in Heavily Treatment-Experienced People with Multidrug-Resistant HIV-1: Week 104 Results of a Phase 2/3 Trial.

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Lenacapavir is a long-acting human immunodeficiency virus type 1 (HIV-1) capsid inhibitor for treatment of HIV-1 infection. We evaluated the efficacy and safety of lenacapavir in combination with an investigator-selected optimized background regimen (OBR) after 104 weeks in adults with multidrug-resistant HIV-1. This ongoing, international, Phase 2/3 trial at 42 sites included 72 adults with multidrug-resistant HIV-1. Following a 2-week oral lenacapavir loading phase, participants received subcutaneous lenacapavir every 26 weeks with an OBR. HIV-1 RNA, CD4 cell counts, and adverse events were assessed over 104 weeks. One participant did not enter the extension phase. At Week 104, 44 of 71 participants (62%, 95% confidence interval [CI]: 50; 73) had HIV-1 RNA <50 copies/mL via US Food and Drug Administration (FDA) snapshot algorithm. When missing data (including discontinuations) were excluded, 44 of 54 participants (82%) had HIV-1 RNA <50 copies/mL at Week 104, mean CD4 cell count increased by 122 cells/ μ L (95% CI: 80; 165), and the proportion of participants with CD4 cell count <200 cells/ μ L decreased from 64% (46 of 72) at Baseline to 29% (16 of 55). Fourteen participants had treatment-emergent lenacapavir resistance; 7 resuppressed (HIV-1 RNA <50 copies/mL) while maintaining lenacapavir use. There were no Grade 4 or serious treatment-related adverse events. One participant discontinued study drug due to an injection site reaction. Treatment with subcutaneous lenacapavir in combination with an OBR was well tolerated and resulted in a high rate of virological suppression over 104 weeks. Lenacapavir represents an important treatment option in people with multidrug-resistant HIV-1.

A Case Report of Lenacapavir Use in a Patient with Multidrug-Resistant HIV: The First Experience in Asia.

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Lenacapavir is a novel, first-in-class, capsid inhibitor, which has been approved as an adjunctive therapy for multidrug-resistant human immunodeficiency virus (HIV)-1 virus in combination with optimized background regimen (OBR). Lenacapavir has demonstrated a significant decrease in viral load and high rate of virologic suppression in patients with multidrug-resistant HIV-1 infection with limited treatment options. Here, we report a case of 43-year-old male who was diagnosed with HIV-1 infection in 2005 but failed to achieve viral suppression due to multiclass resistance. After lenacapavir use with OBR, viral suppression was achieved, and recovery of CD4+ T-cell count was observed for 8 months. This case report shows the first lenacapavir experience in Asia in a heavily treatment-experienced HIV patient with limited treatment options.